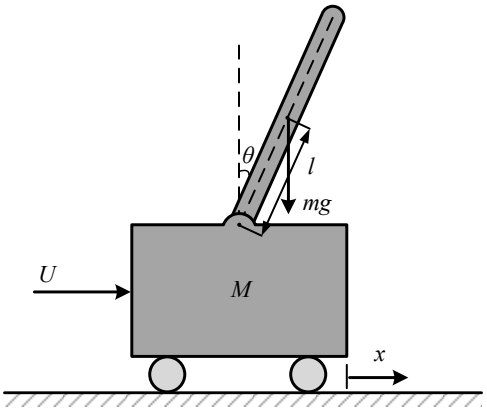
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System Dynamics
Homework 2

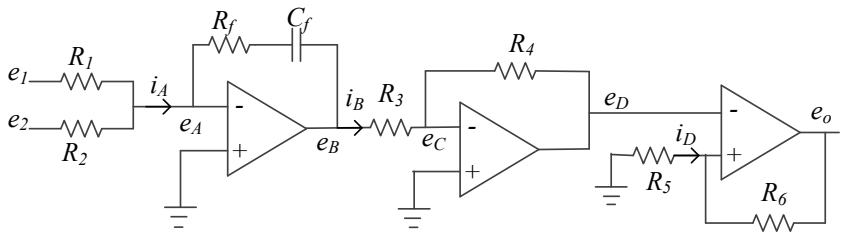
1. Derive the dynamic model of the inverted pendulum;

   Linearize the dynamic model at the equilibrium state ($x = \dot{x} = 0, \theta = \dot{\theta} = 0$) and find the transfer function $\frac{\Theta(s)}{U(s)}$.



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2. Derive the transfer function between E_o and the inputs E_1, $E2$:



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3. For the system shown to the right, derive the transfer function between the input voltage and the rotational speed of J_2:

(a) Draw the free-body diagram

(b) Find $\dfrac{\Omega_2(s)}{E_a(s)}$

